

JEROME FRANCIS

Portfolio: [jerryfrancis-97.github.io](https://github.com/jerryfrancis-97)

francis.jerome999@gmail.com

EDUCATION

T.K.M. College of Engineering, Kollam

August 2016 - 2020

Bachelor of Technology, B.Tech (Hons.)

CGPA : 9.3/10

Department of Computer Science and Engineering(CSE)

Montfort Senior Secondary School, Ashok Vihar (New Delhi)

March 2016

12th Higher Secondary Examination (Science), CBSE

Overall Percentage: 95

10th Secondary Examination, CBSE

CGPA : 9.6/10

EXPERIENCE

Tata Consultancy Services

Oct 2020 - present

Data Analyst, TCS Digital

- Automated tasks to expedite **critical data analysis** for the Data Management team of a leading Investment Bank & Financial Services corporation in NAM.
- Optimized the computation of reports; caused **7x** reduction in TAT for completing process.

Ignitarium Technology Solutions, Kochi

June - July 2019

Machine Learning Intern

- Developed deep learning models for better, **on the fly** fault detection on railway tracks in different terrains.
- Involved in data curation and annotation tasks.
- Developed a new feature that aided the fault detection in railway tracks and was later added into the **Rail Defect Detection Platform**.
- Optimized the model architecture to a lightweight version for embedding into drones.

Radix Info Solutions, New Delhi

July 2018

Summer ML Trainee

- Learned Machine Learning and Deep learning algorithms and implemented them using Tensorflow and Keras.

PROJECTS

Improving IVIM Imaging technique April 2020

In my final-year research project, I worked on different ways to improve **IVIM-MRI imaging technique**. The project investigated methods to reduce the image acquisition time of IVIM MRI technique. Project Supervisor: Dr. Ansamma John, HoD (CSE), TKMCE

Video based RailGap measurement July 2019

Developed a feature to measure rail gaps of railway tracks present in video frames using **Semantic segmentation** and **Image Processing**. This measurement is used to detect gaps which exceed a threshold value (**faulty gaps**). This project was a part of internship done in July 2019.

Smart Waste Management Jan 2019

An IoT based smart waste management system using smart bins to provide waste-management services with optimal amount of time and cost required for the same using machine learning model. Selected for final round of TEKCON'19 contest.[\[Link\]](#)

RailSathi Oct - Dec 2018

Worked in a 3-person team to design a railway booking app as the project for a *Design course*. This application aims to provide booking tickets for connecting trains if direct trains are not possible.

INVOLVEMENT

- **12th position** in MineRL Diamond Challenge (Intro Track) - [NeurIPS '21 competition](#)
- **Student Mentor - AI/ML** Taught basics concepts of ML and AI in SIG-TKMCE club
- **Research Team Member** Worked on drone models in TKMCE Robotics Club
- Qualified for final round in Microsoft AI Challenge, 2018
- Qualified for final round of IdeaFest 2019 for the product 'monitor24'
- Qualified for final round of TEKCON-2019, a state-level expo
- **Volunteer** Contributed 6 hours to Sign Language Workshop as part of TCS empowers, CSR Volunteering Program
- **Volunteer** Conducted the final round of IEEE PES quiz, 2016

ONLINE COURSES AND CERTIFICATIONS

CS285 - Reinforcement Learning, UC Berkley

Reinforcement Learning Specialization by University of Alberta & Amii on Coursera

Learnt about the advanced theoretical concepts of RL and its applications and implementing the concepts which is present in my Github repo [Reinforcement Learning](#)

Game Theory by Stanford University on Coursera

Helped me understand how to structure and analyze different decisions and strategies in simple games. This course covered Nash Equilibrium, Perfect and Imperfect Information Games, Dominant and Trigger Strategies, etc.

Machine Learning by Stanford University on Coursera

Sequence and Models by deeplearning.ai on Coursera

Architecting with Google Compute Engine Specialization by Coursera

Web Development course by UPES ACM chapter

TECHNICAL SKILLS

Languages	Python, C++, C
Tools/Libraries	Keras, Numpy, Pandas, Tensorflow, Pytorch, OpenCV, Scikit-Learn
Simulation	Pybullet, Mujoco
Writing & Editing	L ^A T _E X
Web	Javascript, HTML, CSS